

An Introduction to Optimal Transport Theory

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Summary

This dissertation provides a rigorous introduction to the theory of Optimal Transport. We develop the two principal formulations of the problem, prove the existence of optimal solutions for one of the latter formulation using results from measure theory, and introduce the associated duality theory using tools from optimisation theory. The exposition is built around a unifying example based on electrical power transmission across a network, which is used throughout to give concrete interpretations of the mathematical objects under discussion.

The final chapter turns to the inverse problem: given observed transport data, how can one recover the underlying costs? We review a recent methodology for doing so, present its mathematical formulation in detail, and discuss the assumptions and limitations of the approach.

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Chapter 1

Introduction

From matching supply with demand to reallocating resources across networks, the problem of moving material from one distribution to another in an efficient manner arises in many settings. Such problems require both a description of how mass is distributed and a criterion for the cost of moving it, leading naturally to the mathematical framework of Optimal Transport.

Let X and Y be metric spaces equipped with their Borel σ -algebras and probability measures μ and ν respectively, representing an initial and a target distribution of mass. Given a measurable cost function $c : X \times Y \rightarrow [0, \infty]$, the Optimal Transport problem asks how one may rearrange the mass distributed according to μ into the mass distributed according to ν while minimising the incurred total transport cost. The first formulation we shall study was introduced by the French mathematician Gaspard Monge. The Monge formulation [Monge, 1781] seeks a measurable map $T : X \rightarrow Y$ which transports μ to ν , so that

$$\mathbb{M}(T) := \int_X c(x, T(x)) d\mu(x)$$

is minimised over all such admissible maps. A central difficulty appears immediately: because a transport map sends the mass at each point $x \in X$ to a single destination $T(x) \in Y$, the Monge formulation does not permit splitting mass, and as a result admissible maps may fail to exist.

The Kantorovich formulation, introduced by Leonid Kantorovich [Kantorovich, 1942], addresses this rigidity by replacing transport maps with transport plans, which allow mass to be distributed across several destinations. In this setting, one seeks a probability measure π on $X \times Y$ whose distributions on the first and second coordinates are μ and ν respectively, and which minimises the functional

$$\mathbb{K}(\pi) := \int_{X \times Y} c(x, y) d\pi(x, y).$$

This feature gives Optimal Transport its distinctive character. Because the theory compares distributions by measuring the least cost required to rearrange one into the other, it reflects the geometry of the underlying space rather than relying on pointwise comparison alone. Consequently, it captures structural information that more classical modes of comparison may fail to detect.

Optimal Transport has proved useful in a range of applied settings. In particular, inverse problems provide situations in which observed distributions may be interpreted as the outcome of an efficient redistribution process, making transport based models a natural tool for inferring hidden structure from data. This perspective will be revisited in the final chapter through a discussion of the work of Gaskin et al. [2025].

This dissertation provides a rigorous introduction to Optimal Transport theory for readers familiar with measure theory and optimisation theory at an advanced undergraduate or early postgraduate level. The exposition is centred around a single unifying example. Rather than introducing the theory only through abstract statements and proofs, the main definitions and ideas are developed alongside a motivating model based on electrical power transmission across a network. This example is my own contribution to the exposition and is used throughout to give concrete interpretations of the mathematical objects and results under discussion.

The dissertation is organised as follows. Chapter 2 introduces the basic definitions and notation required throughout, develops the power transmission example, and presents the Monge and Kantorovich formulations of Optimal Transport together with their relationship. Chapter 3 proves the existence of minimisers for the Kantorovich problem under light restrictions on the cost function. Chapter 4 develops Kantorovich duality. The final chapter reviews the inverse optimal transport methodology of Gaskin et al. [2025], in which observed transport plans are used to recover underlying cost structures. We present its mathematical formulation in detail and conclude with a discussion of the method's assumptions and limitations.

For each proof we note the extent to which it follows existing sources. If no such note is provided the argument may be assumed to be our own. This indicates only that the proof in this dissertation is ours, not that the proof is original. It may appear elsewhere or follow a standard approach, but we present it in our own words and our own way.

Chapter 2

The Monge and Kantorovich Formulations of Optimal Transport

This chapter develops the two principal formulations of the Optimal Transport problem and explains the relationship between them. It begins by introducing the basic definitions and notation required throughout, after which the power transmission example is presented to guide the discussion. The Monge formulation is studied first, since it provides the most direct expression of the idea of transporting mass by a map. This is followed by the introduction of transport plans and the Kantorovich formulation. The chapter concludes by showing a connection between the Monge and Kantorovich formulations.

2.1 Basic definitions and notation

The definitions and notations in this section are found in Thorpe [2023, Chapter 1] unless otherwise indicated.

We denote by $\mathcal{P}(X)$ the set of all probability measures on $(X, \mathcal{B}(X))$.

We denote by $\mathcal{C}_b^0(X)$ the space of all bounded continuous functions on X .

Definition 2.1. A Polish space is a separable completely metrizable topological space.

Definition 2.2. A measure μ defined on $(X, \mathcal{B}(X))$ is called a Borel measure.

Definition 2.3 ([Bogachev, 2007a, Section 1.3]). For $x \in X$, the Dirac measure at x is the probability measure δ_x on $(X, \mathcal{B}(X))$ defined by

$$\delta_x(A) = \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{if } x \notin A, \end{cases}$$

for every $A \in \mathcal{B}(X)$. In particular, for any $f \in \mathcal{C}_b^0(X)$,

$$\int_X f d\delta_x = f(x).$$

Definition 2.4. For a measure μ on $(X, \mathcal{B}(X))$, we denote by $L^1(\mu)$ the space of all measurable functions $f : X \rightarrow \mathbb{R}$ satisfying

$$\int_X |f(x)| d\mu(x) < \infty.$$

Throughout the dissertation, we shall assume these spaces are equipped with their Borel σ -algebras, $\mathcal{B}(X)$. From this point onward, when referring to a probability measure μ on X , what we mean is that μ is on $(X, \mathcal{B}(X))$.

We denote by $\mathcal{M}_+(X)$ the set of all positive Borel measures on X .

2.2 The power transmission example

As a motivating example, we consider the transmission of electrical power across a network. Power stations and households are distributed across a region, and electricity must be sent from points of supply to points of demand. Ignoring transmission losses, we take the objective to be the minimisation of transport cost, which we interpret in terms of distance travelled. Henceforth, the transmission of electricity will be viewed as the movement of mass, with source and target distributions represented by probability measures μ and ν and the cost of moving one unit of mass from x to y given by a cost function $c : X \times Y \rightarrow [0, +\infty]$.

2.3 Transport Maps and the Monge Formulation

Having introduced the power transmission example and the cost function, we require a mathematical object capable of describing how mass is moved from the source space to the target space. The most direct approach is to assign, for each point $x \in X$, a single destination $T(x) \in Y$. In the context of the power transmission model, this corresponds to assigning the mass located at each source point to one point of demand. Although this is not realistic for an actual power grid, it gives a simple and tangible illustration of the rigidity of the Monge formulation. This leads naturally to the notion of a transport map.

Such a map should satisfy two requirements. First, it must preserve mass: for every Borel set $B \subseteq Y$, the total mass arriving in B should equal the total mass initially located at points that are sent into B . Through the lens of our example, the electricity flowing into a household must be equivalent to how much was originally stored in

the power stations that supplied power to that house. It also enforces that one power station may only supply power to one house.

Second, it must be measurable, so that these transported masses are well-defined. These conditions motivate the following definition.

Definition 2.5 (Thorpe, 2023, Definition 2.1). Take $\mu \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Y)$. A measurable map $T : X \rightarrow Y$ is said to transport μ to ν if

$$\nu(B) = \mu(T^{-1}(B))$$

for all measurable sets $B \subseteq Y$. If this condition holds, we write $T_{\#}\mu = \nu$.

To provide an explicit example of such a case where a transport map may fail to exist, we show the following example taken verbatim from Thorpe [2023, Chapter 1]. Define $\mu = \delta_x$ and $\nu = \frac{1}{2}\delta_{y_1} + \frac{1}{2}\delta_{y_2}$ where $y_1 \neq y_2$. Then $\nu(\{y_1\}) = \frac{1}{2}$, but $\mu(T^{-1}(\{y_1\}))$ can only be 0 or 1, depending on whether the mass of μ is sent to y_1 or y_2 . Hence, no valid transport map exists.

Returning to our original focus, once a cost function $c : X \times Y \rightarrow [0, \infty]$ has been fixed, the aim is to find a transport map that moves μ to ν while minimising the total transport cost. This leads to the Monge formulation of Optimal Transport.

Definition 2.6 (Monge Formulation, Thorpe, 2023, Definition 2.2).

Take $\mu \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Y)$. The Monge problem is defined by

$$\inf_{T : T_{\#}\mu = \nu} \mathbb{M}(T), \quad \mathbb{M}(T) := \int_X c(x, T(x)) d\mu(x),$$

where the infimum is taken over all admissible transport maps $T : X \rightarrow Y$.

The Monge formulation is conceptually simple, since it seeks an optimal rule for moving mass from source to target. However, we have seen that its simplicity comes at a cost. Namely, the problem may be infeasible. This difficulty motivates the more flexible Kantorovich formulation, which will be introduced later in the chapter.

2.4 Transport plans

Having discussed the main limitation of transport maps, we will now discuss their more flexible counterparts: transport plans.

Definition 2.7 (Villani, 2003, Introduction).

A transport plan between $\mu \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Y)$ is a probability measure $\pi \in \mathcal{P}(X \times Y)$ such that

$$\begin{aligned}\pi(A \times Y) &= \mu(A), \\ \pi(X \times B) &= \nu(B),\end{aligned}$$

for all measurable sets $A \subseteq X$ and $B \subseteq Y$. In other words, π has first marginal μ and second marginal ν .

Alternatively, this may be written in terms of the coordinate projections

$$P^X : X \times Y \rightarrow X, \quad P^Y : X \times Y \rightarrow Y,$$

defined by

$$P^X(x, y) = x, \quad P^Y(x, y) = y.$$

Then the marginal constraints become

$$\begin{aligned}(P^X)_\# \pi &= \mu, \\ (P^Y)_\# \pi &= \nu.\end{aligned}$$

Equivalently, for $\phi \in \mathcal{C}_b^0(X)$ and $\psi \in \mathcal{C}_b^0(Y)$,

$$\begin{aligned}\int_{X \times Y} \phi(x) d\pi(x, y) &= \int_X \phi(x) d\mu(x), \\ \int_{X \times Y} \psi(y) d\pi(x, y) &= \int_Y \psi(y) d\nu(y).\end{aligned}$$

Interpreting this once more through the lens of our original example, we can view this as allowing the electricity generated at any power station to be transmitted to many households. This is a more realistic modelling assumption illustrating the added flexibility that transport plans provide.

Definition 2.8. We denote the set of all transport plans with first marginal μ and second marginal ν as $\Pi(\mu, \nu)$.

We give a direct proof of the following standard result. The construction used for non-emptiness follows Thorpe [2023, Section 2.6].

Lemma 2.9. $\Pi(\mu, \nu)$ is non-empty and convex.

Proof. We first show that $\Pi(\mu, \nu)$ is non-empty.

Consider the product measure $\pi = \mu \otimes \nu$, defined by $(\mu \otimes \nu)(A \times B) = \mu(A)\nu(B)$.

We observe that

$$\begin{aligned} (\mu \otimes \nu)(A \times Y) &= \mu(A) \cdot \nu(Y) = \mu(A) \cdot 1 = \mu(A), \\ (\mu \otimes \nu)(X \times B) &= \mu(X) \cdot \nu(B) = 1 \cdot \nu(B) = \nu(B). \end{aligned}$$

Hence $\mu \otimes \nu \in \Pi(\mu, \nu)$, and therefore $\Pi(\mu, \nu)$ is non-empty.

We now show that $\Pi(\mu, \nu)$ is convex.

Take $\pi_1, \pi_2 \in \Pi(\mu, \nu)$ and let $t \in [0, 1]$. Define $\pi := t\pi_1 + (1-t)\pi_2$, which is a valid probability measure as it is clearly non-negative and has unit mass:

$$\pi(X \times Y) = t\pi_1(X \times Y) + (1-t)\pi_2(X \times Y) = t(1) + (1-t)(1) = 1.$$

Next, we verify that π has first marginal μ and second marginal ν .

For the first marginal, we have

$$\begin{aligned} \pi(A \times Y) &= (t\pi_1 + (1-t)\pi_2)(A \times Y) \\ &= t\pi_1(A \times Y) + (1-t)\pi_2(A \times Y) \\ &= t\mu(A) + (1-t)\mu(A) \\ &= \mu(A). \end{aligned}$$

Similarly, for the second marginal,

$$\begin{aligned} \pi(X \times B) &= (t\pi_1 + (1-t)\pi_2)(X \times B) \\ &= t\pi_1(X \times B) + (1-t)\pi_2(X \times B) \\ &= t\nu(B) + (1-t)\nu(B) \\ &= \nu(B). \end{aligned}$$

Therefore $\pi \in \Pi(\mu, \nu)$. Since every convex combination of two elements of $\Pi(\mu, \nu)$ remains in $\Pi(\mu, \nu)$, it follows that $\Pi(\mu, \nu)$ is convex. $\square$

2.5 The Kantorovich formulation

We have now established the necessary theory to state the Kantorovich problem:

Definition 2.10 (Kantorovich formulation, Thorpe, 2023, Definition 2.3).

Given $\mu \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Y)$, the Kantorovich problem is defined by

$$\inf_{\pi \in \Pi(\mu, \nu)} \mathbb{K}(\pi), \quad \mathbb{K}(\pi) := \int_{X \times Y} c(x, y) d\pi(x, y),$$

where we may think of $d\pi(x, y)$ as the mass we transfer from position x to y expressed as a proportion of the total mass present.

The following result is noted in Thorpe [2023, Chapter 2]. We provide a full proof.

Theorem 2.11. *The Kantorovich Problem is a convex optimisation problem.*

Proof. By Lemma 2.9, the feasible set $\Pi(\mu, \nu)$ is non-empty and convex. It therefore remains only to show that the objective functional $\mathbb{K}$ is convex in π .

Take $\pi_1, \pi_2 \in \Pi(\mu, \nu)$ and let $t \in [0, 1]$. Then we have

$$\begin{aligned} \mathbb{K}(t\pi_1 + (1-t)\pi_2) &= \int_{X \times Y} c(x, y) d(t\pi_1 + (1-t)\pi_2) \\ &= t \cdot \int_{X \times Y} c(x, y) d\pi_1 + (1-t) \cdot \int_{X \times Y} c(x, y) d\pi_2 \\ &= t \cdot \mathbb{K}(\pi_1) + (1-t) \cdot \mathbb{K}(\pi_2). \end{aligned}$$

In fact, what we have shown is that the objective functional is linear in π , and is therefore also convex in π . Since the feasible set is convex and the objective functional is linear, hence convex, the Kantorovich problem is a convex minimisation problem. $\square$

This is a particularly significant result, since convex optimisation is a rich and extensively studied field, and its theory provides a range of powerful tools that can be applied to the Kantorovich formulation. Furthermore, this perspective leads naturally to a dual formulation, which can be used to obtain lower bounds for the primal problem and will be discussed in Chapter 4. In the case of discrete measures, the Kantorovich formulation reduces to a linear programme, which will be explored in Chapter 5.

2.6 Linking the Monge and Kantorovich formulations

The following result appears in Thorpe [2023, Chapter 2]. We present a proof in greater detail than what is shown there.

Theorem 2.12. *Every admissible Monge transport map induces a Kantorovich transport plan with the same cost. Consequently,*

$$\inf_{\pi \in \Pi(\mu, \nu)} \mathbb{K}(\pi) \leq \inf_{T: T_{\#}\mu = \nu} \mathbb{M}(T).$$

Although this may seem a simple result, a rigorous proof requires some work.

Proof. Let $T : X \rightarrow Y$ be an admissible transport map for the Monge problem, so that for every ν -measurable set $B \subseteq Y$, $\nu(B) = \mu(T^{-1}(B))$.

Define a map $F : X \rightarrow X \times Y$ by $F(x) = (x, T(x))$, and let $\pi := F_{\#}\mu$. Thus, for every measurable set $E \subseteq X \times Y$, $\pi(E) = \mu(F^{-1}(E))$.

We now verify that π is a transport plan with marginals μ and ν . For the first marginal, let $A \subseteq X$ be measurable. Then

$$\begin{aligned}\pi(A \times Y) &= \mu(F^{-1}(A \times Y)) \\ &= \mu(\{x \in X : (x, T(x)) \in A \times Y\}).\end{aligned}$$

But $(x, T(x)) \in A \times Y$ if and only if $x \in A$, and so $F^{-1}(A \times Y) = A$. Therefore, $\pi(A \times Y) = \mu(A)$, showing that the first marginal of π is μ .

For the second marginal, let $B \subseteq Y$ be measurable. Then

$$\begin{aligned}\pi(X \times B) &= \mu(F^{-1}(X \times B)) \\ &= \mu(\{x \in X : (x, T(x)) \in X \times B\}).\end{aligned}$$

Now $(x, T(x)) \in X \times B$ if and only if $x \in T^{-1}(B)$, and hence $F^{-1}(X \times B) = T^{-1}(B)$.

It follows that $\pi(X \times B) = \mu(T^{-1}(B)) = \nu(B)$, so the second marginal of π is ν . Hence $\pi \in \Pi(\mu, \nu)$.

We now compare the costs. Since $\pi = F_{\#}\mu$, we obtain

$$\mathbb{K}(\pi) = \int_{X \times Y} c(x, y) d\pi(x, y) = \int_X c(F(x)) d\mu(x) = \int_X c(x, T(x)) d\mu(x) = \mathbb{M}(T).$$

Therefore, every admissible Monge transport map gives rise to a Kantorovich transport plan with exactly the same cost. Taking infima over all admissible transport maps yields

$$\inf_{\pi \in \Pi(\mu, \nu)} \mathbb{K}(\pi) \leq \inf_{T: T_{\#}\mu = \nu} \mathbb{M}(T).$$

□

Chapter 3

Existence of Minimisers for the Kantorovich Formulation

This chapter proves the existence of minimisers for the Kantorovich problem under the assumption that the cost function is lower semi-continuous and has function values restricted to $[0, +\infty)$. We first state the definition of lower semi-continuity.

Definition 3.1 (Bogachev, 2007a, Section 2.12). A function $f : X \rightarrow \mathbb{R} \cup \{-\infty, +\infty\}$ is lower semi-continuous if for every $c \in [-\infty, \infty]$, the set $\{x : f(x) > c\}$ is open.

We now state the theorem this chapter will be dedicated to proving. It can be found in Thorpe [2023, Proposition 2.4].

Theorem 3.2. *Let $\mu \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Y)$, where X and Y are Polish spaces, and let $c : X \times Y \rightarrow [0, \infty)$ be lower semi-continuous. Then there exists $\pi^* \in \Pi(\mu, \nu)$ such that*

$$\mathbb{K}(\pi^*) \leq \mathbb{K}(\pi) \quad \text{for all } \pi \in \Pi(\mu, \nu).$$

As stated in Thorpe [2023, Chapter 2], the key idea behind the proof is to first show that the admissible set $\Pi(\mu, \nu)$ is compact in the weak topology. One then takes a minimising sequence $(\pi_n)_{n \in \mathbb{N}} \subset \Pi(\mu, \nu)$ whose costs converge to the infimum of the Kantorovich problem. By compactness, this sequence admits a weakly convergent subsequence with limit $\pi^* \in \Pi(\mu, \nu)$. The lower semi-continuity of the cost function then yields $\mathbb{K}(\pi^*) \leq \liminf_{n \rightarrow \infty} \mathbb{K}(\pi_n)$, by a theorem that will be stated in the next section, from which it follows that π^* is a minimiser.

We will therefore complete a rigorous proof in two stages. We initially show that $\Pi(\mu, \nu)$ is compact in the weak topology, and then complete the proof using the sequence argument above. To make this argument rigorous, we must first establish several preliminary definitions and results, which will be given in the next section.

3.1 Key Definitions

The following definitions assume X to be any topological space.

Definition 3.3 (Bogachev, 2007b, Definition 7.1.1).

A Borel measure μ on X is called a Radon measure if for every $B \in \mathcal{B}(X)$ and $\varepsilon > 0$, there exists a compact set $K_\varepsilon \subset B$ such that $\mu(B \setminus K_\varepsilon) < \varepsilon$.

Definition 3.4 (Bogachev, 2007b, Definition 8.6.1).

A family of Radon measures on X is said to be uniformly tight if for every $\varepsilon > 0$, there exists a compact set K_ε such that $\mu(X \setminus K_\varepsilon) < \varepsilon$ for all measures μ in the family.

Definition 3.5 (Thorpe, 2023, Chapter 1).

Let $(\mu_n)_{n \in \mathbb{N}} \subset \mathcal{P}(X)$ be a sequence of probability measures.

We say that μ_n converges weakly to μ if

$$\int_X f(x) d\mu_n(x) \xrightarrow{n \rightarrow \infty} \int_X f(x) d\mu(x)$$

for all $f \in C_b^0(X)$.

Definition 3.6 (Billingsley, 1999, Section 5).

A set Π of probability measures is said to be relatively compact if for every sequence of probability measures $(P_n)_{n \in \mathbb{N}} \subset \Pi$, there exists a subsequence $(P_{n_i})_{i \in \mathbb{N}}$ and a probability measure Q such that $P_{n_i} \xrightarrow{i \rightarrow \infty} Q$.

3.2 Key Results

Lemma 3.7 (Bogachev, 2007b, Theorem 7.1.7). *If X is a complete and separable metric space, then every Borel measure μ on X is Radon.*

A direct consequence of this theorem is that any Borel measure on a Polish space is guaranteed to be Radon, as Polish metric spaces are by definition complete and separable.

Theorem 3.8 (Prokhorov's Theorem, Billingsley, 1999, Section 5). *A family of probability measures Π on a separable and complete metric space is relatively compact if and only if it is tight.*

Theorem 3.9 (Portmanteau Theorem, partial statement, Thorpe, 2023, Theorem 1.2).

We record only the part of the Portmanteau theorem that will be needed for our purposes. Let X be a metric space, let $(\mu_n)_{n \in \mathbb{N}} \subset \mathcal{P}(X)$, and let $\mu \in \mathcal{P}(X)$. If μ_n converges weakly to μ , then

$$\liminf_{n \rightarrow \infty} \int_X f(x) d\mu_n(x) \geq \int_X f(x) d\mu(x)$$

for every lower semi-continuous function f bounded from below.

3.3 Proof of Theorem 3.2

We are now ready to prove Theorem 3.2. The entire proof follows the structure of the proof given in Thorpe [2023, Chapter 2], but is presented in greater detail. We begin with the following compactness lemma.

Lemma 3.10. *The set $\Pi(\mu, \nu)$ is compact in the weak topology on $\mathcal{P}(X \times Y)$.*

Proof. We begin by showing that $\Pi(\mu, \nu)$ is relatively compact.

Let $\varepsilon > 0$. Choose compact sets $K \subset X$ and $L \subset Y$ such that

$$\mu(X \setminus K) < \frac{\varepsilon}{2} \quad \text{and} \quad \nu(Y \setminus L) < \frac{\varepsilon}{2}.$$

The existence of such sets is guaranteed since μ and ν are defined on Polish spaces and are therefore Radon measures by Lemma 3.7.

Now observe that if $(x, y) \in (X \times Y) \setminus (K \times L)$, then it is not possible to have both $x \in K$ and $y \in L$ simultaneously. Hence

$$(X \times Y) \setminus (K \times L) \subseteq ((X \setminus K) \times Y) \cup (X \times (Y \setminus L)).$$

$$\begin{aligned} \text{Therefore, } \pi((X \times Y) \setminus (K \times L)) &\leq \pi((X \setminus K) \times Y) + \pi(X \times (Y \setminus L)) \\ &= \mu(X \setminus K) + \nu(Y \setminus L) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Hence $\Pi(\mu, \nu)$ is tight, and so by Prokhorov's Theorem it is relatively compact.

To prove compactness, it remains to show that $\Pi(\mu, \nu)$ is closed.

Take a sequence $(\pi_n)_{n \in \mathbb{N}} \subset \Pi(\mu, \nu)$ such that π_n converges weakly to some probability measure $\pi \in \mathcal{P}(X \times Y)$. If we can show that π has marginals μ and ν , then it will follow that $\pi \in \Pi(\mu, \nu)$.

By the definition of weak convergence, we know that

$$\int_{X \times Y} f(x, y) d\pi_n(x, y) \xrightarrow{n \rightarrow \infty} \int_{X \times Y} f(x, y) d\pi(x, y)$$

for all $f \in \mathcal{C}_b^0(X \times Y)$.

Now define $f(x, y) = g(x)$, where $g \in \mathcal{C}_b^0(X)$.

Then

$$\int_{X \times Y} f(x, y) d\pi_n(x, y) = \int_{X \times Y} g(x) d\pi_n(x, y) = \int_X g(x) d\mu(x),$$

since π_n has first marginal μ .

We also note that

$$\int_{X \times Y} g(x) d\pi(x, y) = \int_X g(x) d(P_{\#}^X \pi)(x).$$

Using what we have assembled thus far, we may conclude

$$\int_X g(x) d\mu(x) = \int_X g(x) d(P_{\#}^X \pi)(x).$$

As this holds for all $g \in C_b^0(X)$, we conclude that $P_{\#}^X \pi = \mu$, and therefore π has first marginal μ . An almost identical argument shows that $P_{\#}^Y \pi = \nu$, so that π has second marginal ν . Hence $\pi \in \Pi(\mu, \nu)$.

Since $\Pi(\mu, \nu)$ is both relatively compact and closed, every sequence of probability measures in $\Pi(\mu, \nu)$ admits a weakly convergent subsequence whose limit remains in $\Pi(\mu, \nu)$. Therefore $\Pi(\mu, \nu)$ is compact. $\square$

We now complete the proof of Theorem 3.2.

Proof. We begin by defining $m := \inf_{\pi \in \Pi(\mu, \nu)} \mathbb{K}(\pi)$, which is guaranteed to exist since $\mathbb{K}(\pi)$ is bounded below by 0 and $\Pi(\mu, \nu)$ is non-empty.

By the definition of the infimum, we may choose a sequence of probability measures $(\pi_n)_{n \in \mathbb{N}} \subset \Pi(\mu, \nu)$ such that $\mathbb{K}(\pi_n) \xrightarrow{n \rightarrow \infty} m$.

Since $\Pi(\mu, \nu)$ is compact by Lemma 3.10, there exists a subsequence $(\pi_{n_i})_{i \in \mathbb{N}}$ and a probability measure $\pi^* \in \Pi(\mu, \nu)$ such that $\pi_{n_i} \xrightarrow{i \rightarrow \infty} \pi^*$, which we claim to be our minimiser.

As $(\pi_{n_i})_{i \in \mathbb{N}}$ converges weakly to π^* , and c is lower semi-continuous and bounded below by 0, the Portmanteau Theorem yields

$$\lim_{i \rightarrow \infty} \mathbb{K}(\pi_{n_i}) = \liminf_{i \rightarrow \infty} \int_{X \times Y} c(x, y) d\pi_{n_i}(x, y) \geq \int_{X \times Y} c(x, y) d\pi^*(x, y) = \mathbb{K}(\pi^*).$$

As $(\pi_n)_{n \in \mathbb{N}}$ is a minimising sequence and $(\pi_{n_i})_{i \in \mathbb{N}}$ is a subsequence, it follows that $(\pi_{n_i})_{i \in \mathbb{N}}$ is also minimising. Therefore, $\lim_{i \rightarrow \infty} \mathbb{K}(\pi_{n_i}) = m$.

Thus the inequality above becomes $\mathbb{K}(\pi^*) \leq m$.

On the other hand, since $\pi^* \in \Pi(\mu, \nu)$, the definition of the infimum gives $\mathbb{K}(\pi^*) \geq m$.

Hence $\mathbb{K}(\pi^*) = m$, and therefore π^* is a minimiser. $\square$

Chapter 4

Kantorovich Duality

In this chapter, we introduce the dual problem associated with the Kantorovich formulation. From optimisation theory, we recall that the dual problem is a concave maximisation problem constructed to provide a lower bound for the value of a minimisation problem. We additionally recall that if the supremum of the dual problem is less than or equal to the infimum of the primal problem, weak duality is said to hold. If equality holds, then strong duality is said to hold.

We begin by independently deriving the dual function, after which weak duality is proved directly. We then outline the main ideas used to establish strong duality, giving a skeleton proof, rather than a full technical argument which can be found in Villani [2003, Theorem 1.3]. We note that weak duality is guaranteed to hold if strong duality holds, but as we provide only a skeleton proof, we prove weak duality outright. The chapter concludes with an interpretation of the result and its significance within Optimal Transport theory.

Throughout this chapter, we will assume X and Y to be Polish spaces, $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$, and $c : X \times Y \rightarrow [0, +\infty]$.

4.1 Deriving the Lagrangian Dual

We begin by recalling the primal Kantorovich problem given in Definition 2.10. From the definition of a transport plan, we obtain the equality constraints

$$\begin{aligned}(P^X)_\# \pi - \mu &= 0, \\ (P^Y)_\# \pi - \nu &= 0.\end{aligned}$$

In deriving the dual function, we temporarily optimise over positive Borel measures $\pi \in \mathcal{M}_+(X \times Y)$, while enforcing the marginal constraints through Lagrange multipliers. This is standard practice when deriving the dual.

4.1.1 Forming the Lagrangian

We take $f \in \mathcal{C}_b^0(X)$ and $g \in \mathcal{C}_b^0(Y)$ as our Lagrange multipliers. Since we are optimising over measures, the corresponding multipliers are functions rather than scalar constants.

The Lagrangian is the function $\mathcal{L} : \mathcal{M}_+(X \times Y) \times \mathcal{C}_b^0(X) \times \mathcal{C}_b^0(Y) \rightarrow \mathbb{R}$ given by

$$\mathcal{L}(\pi, f, g) = \int_{X \times Y} c(x, y) d\pi(x, y) + \int_X f(x) d(\mu - (P^X)_\# \pi) + \int_Y g(y) d(\nu - (P^Y)_\# \pi).$$

Expanding and collecting like terms yields

$$\mathcal{L}(\pi, f, g) = \int_X f(x) d\mu(x) + \int_Y g(y) d\nu(y) + \int_{X \times Y} (c(x, y) - f(x) - g(y)) d\pi(x, y).$$

4.1.2 Forming the Dual

We now form the dual by minimising $\mathcal{L}(\pi, f, g)$ over $\mathcal{M}_+(X \times Y)$. Thus,

$$\begin{aligned} \mathbb{J}(f, g) &:= \inf_{\pi \geq 0} \mathcal{L}(\pi, f, g) \\ &= \inf_{\pi \geq 0} \left[\int_X f(x) d\mu(x) + \int_Y g(y) d\nu(y) + \int_{X \times Y} (c(x, y) - f(x) - g(y)) d\pi(x, y) \right] \\ &= \int_X f(x) d\mu(x) + \int_Y g(y) d\nu(y) + \inf_{\pi \geq 0} \int_{X \times Y} (c(x, y) - f(x) - g(y)) d\pi(x, y). \end{aligned}$$

To evaluate $\inf_{\pi \geq 0} \int_{X \times Y} (c(x, y) - f(x) - g(y)) d\pi(x, y)$ we examine two cases.

Case 1. Suppose $f(x) + g(y) \leq c(x, y)$ for all $(x, y) \in X \times Y$, which yields

$$\int_{X \times Y} (c(x, y) - f(x) - g(y)) d\pi(x, y) \geq 0$$

for every positive measure π . Taking π to be the zero measure, we see that

$$\inf_{\pi \geq 0} \int_{X \times Y} (c(x, y) - f(x) - g(y)) d\pi(x, y) = 0.$$

Case 2. We handle this case in the manner shown in Thorpe [2023, Section 4.1].

Suppose there exists $(x, y) \in X \times Y$ such that $c(x, y) - f(x) - g(y) = \varepsilon < 0$.

Then by defining $\pi = \lambda \delta_{(x, y)}$ for $\lambda > 0$, we find that

$$\inf_{\pi \geq 0} \int_{X \times Y} (c(x, y) - f(x) - g(y)) d\pi(x, y) \leq \lambda \varepsilon \xrightarrow{\lambda \rightarrow \infty} -\infty.$$

It follows that the dual function is given by

$$\mathbb{J}(f, g) := \begin{cases} \int_X f(x) d\mu(x) + \int_Y g(y) d\nu(y), & \text{if } c(x, y) \geq f(x) + g(y) \text{ for all } (x, y) \in X \times Y, \\ -\infty, & \text{otherwise.} \end{cases}$$

We therefore define the Lagrangian dual problem as

$$\sup_{f,g \in \Phi_c} \mathbb{J}(f,g) = \sup_{f,g \in \Phi_c} \left\{ \int_X f(x) d\mu(x) + \int_Y g(y) d\nu(y) \right\},$$

where

$$\Phi_c = \{(f,g) \in L^1(\mu) \times L^1(\nu) : f + g \leq c\}.$$

4.2 Proving Weak Duality

Theorem 4.1 (Weak duality). *Let $\inf_{\pi \in \Pi(\mu,\nu)} \int_{X \times Y} c(x,y) d\pi(x,y)$ be the Kantorovich primal problem, and let $\sup_{(\phi,\psi) \in \Phi_c} \left\{ \int_X \phi(x) d\mu(x) + \int_Y \psi(y) d\nu(y) \right\}$ be its dual problem.*

Then for every primal feasible $\pi \in \Pi(\mu,\nu)$ and every dual feasible pair (ϕ,ψ) , we have

$$\int_X \phi(x) d\mu(x) + \int_Y \psi(y) d\nu(y) \leq \int_{X \times Y} c(x,y) d\pi(x,y).$$

Consequently,

$$\sup_{(\phi,\psi) \in \Phi_c} \mathbb{J}(\phi,\psi) \leq \inf_{\pi \in \Pi(\mu,\nu)} \mathbb{K}(\pi).$$

Proof. Let $\pi \in \Pi(\mu,\nu)$ be primal feasible, and let (ϕ,ψ) be dual feasible. Since (ϕ,ψ) is dual feasible, we have $\phi(x) + \psi(y) \leq c(x,y)$ for all $(x,y) \in X \times Y$.

Integrating both sides with respect to π yields

$$\int_{X \times Y} (\phi(x) + \psi(y)) d\pi(x,y) \leq \int_{X \times Y} c(x,y) d\pi(x,y).$$

Since π has marginals μ and ν , we obtain

$$\int_{X \times Y} \phi(x) d\pi(x,y) = \int_X \phi(x) d\mu(x),$$

and

$$\int_{X \times Y} \psi(y) d\pi(x,y) = \int_Y \psi(y) d\nu(y).$$

Therefore,

$$\int_X \phi(x) d\mu(x) + \int_Y \psi(y) d\nu(y) \leq \int_{X \times Y} c(x,y) d\pi(x,y).$$

Since this holds for every primal feasible π and every dual feasible pair (ϕ,ψ) , taking the supremum over all dual feasible pairs on the left and the infimum over all primal feasible transport plans on the right gives

$$\sup_{(\phi,\psi) \in \Phi_c} \left\{ \int_X \phi d\mu + \int_Y \psi d\nu \right\} \leq \inf_{\pi \in \Pi(\mu,\nu)} \int_{X \times Y} c d\pi.$$

□

4.3 Outline of the proof of Strong Duality

This section is concerned with sketching a proof of strong duality, namely that

$$\sup_{(\phi, \psi) \in \Phi_c} \mathbb{J}(\phi, \psi) = \inf_{\pi \in \Pi(\mu, \nu)} \mathbb{K}(\pi).$$

A skeleton proof will be provided, following what has been provided in Thorpe [2023, Section 4.1]. The step requiring more careful justification will be indicated explicitly.

Theorem 4.2 (Strong duality, proof sketch). *Let $\mathbb{K}$ and $\mathbb{J}$ be the Kantorovich primal and dual functions respectively. We claim that strong duality holds, yielding*

$$\sup_{(\phi, \psi) \in \Phi_c} \mathbb{J}(\phi, \psi) = \inf_{\pi \in \Pi(\mu, \nu)} \mathbb{K}(\pi).$$

Proof. We begin by showing that $\inf_{\pi \in \Pi(\mu, \nu)} \mathbb{K}(\pi) = \inf_{\pi \geq 0} \sup_{\phi, \psi} \mathcal{L}(\pi, \phi, \psi)$.

Recall that

$$\mathcal{L}(\pi, \phi, \psi) = \int_{X \times Y} c \, d\pi + \int_X \phi \, d(\mu - (P^X)_\# \pi) + \int_Y \psi \, d(\nu - (P^Y)_\# \pi).$$

Hence,

$$\sup_{\phi, \psi} \mathcal{L}(\pi, \phi, \psi) = \int_{X \times Y} c \, d\pi + \sup_{\phi} \int_X \phi \, d(\mu - (P^X)_\# \pi) + \sup_{\psi} \int_Y \psi \, d(\nu - (P^Y)_\# \pi).$$

Examining $\int_X \phi(x) \, d(\mu - (P^X)_\# \pi)$ in more detail, we note if $\mu \neq (P^X)_\# \pi$, then there exists a measurable set A such that $(\mu - (P^X)_\# \pi)(A) \neq 0$.

Defining $\phi = \lambda \mathbf{1}_A$, we obtain

$$\int_X \phi(x) \, d(\mu - (P^X)_\# \pi)(x) = \lambda(\mu - (P^X)_\# \pi)(A).$$

Then, without loss of generality, if $(\mu - (P^X)_\# \pi)(A) > 0$, letting $\lambda \rightarrow \infty$ forces

$$\sup_{\phi} \int_X \phi(x) \, d(\mu - (P^X)_\# \pi)(x) = +\infty.$$

Hence, we require $\mu = (P^X)_\# \pi$ in order for $\sup_{\phi, \psi} \mathcal{L}(\pi, \phi, \psi)$ to remain finite. By an almost identical argument, we must also have $\nu = (P^Y)_\# \pi$.

Thus π must be a transport plan with marginals μ and ν , that is, $\pi \in \Pi(\mu, \nu)$, and $\sup_{\phi, \psi} \mathcal{L}(\pi, \phi, \psi)$ reduces to $\int_{X \times Y} c(x, y) \, d\pi(x, y)$. It follows that

$$\inf_{\pi \geq 0} \sup_{\phi, \psi} \mathcal{L}(\pi, \phi, \psi) = \inf_{\pi \in \Pi(\mu, \nu)} \int_{X \times Y} c(x, y) \, d\pi(x, y),$$

which is precisely the Kantorovich primal problem.

Now, assuming the existence of a minimax principle that allows us to write

$$\inf_{\pi \geq 0} \sup_{\phi, \psi} \mathcal{L}(\pi, \phi, \psi) = \sup_{\phi, \psi} \inf_{\pi \geq 0} \mathcal{L}(\pi, \phi, \psi),$$

we may complete the proof by identifying

$$\sup_{\phi, \psi} \inf_{\pi \geq 0} \mathcal{L}(\pi, \phi, \psi)$$

as the optimal dual value. We do not prove the existence of the minimax principle, as this requires significantly more work and advanced results from convex analysis and algebraic topology that are beyond the scope of this exposition. A rigorous proof can be found in Villani [2003, Theorem 1.3]. $\square$

4.4 Interpretation of Kantorovich Duality

This section provides an intuitive interpretation of Kantorovich Duality. We do this by recasting the interpretation given in Villani [2003, Section 1.1.3] through the lens of our original motivating example. We briefly recall that our objective was to minimise the overall cost of transmitting electricity across the power network.

Now imagine that, instead of paying directly for transmission along the network, an operator offers a pricing scheme. For each unit of electricity collected at a power station located at x , you pay a loading price $\phi(x)$, and for each unit delivered to a household located at y , you pay an unloading price $\psi(y)$. To ensure that this scheme never costs more than direct transmission, the operator chooses these prices so that

$$\phi(x) + \psi(y) \leq c(x, y)$$

for every pair (x, y) . In other words, the combined loading and unloading charge for moving one unit of electricity from x to y never exceeds the actual transmission cost. Kantorovich duality states that one can choose ϕ and ψ so that the total value of this pricing scheme is exactly equal to the minimum cost of transporting the electricity itself.

This interpretation explains how the dual problem serves as an alternative description of the same transport problem.

Chapter 5

Inverse Optimal Transport

In the preceding chapters, we studied the forward optimal transport problem: given a cost function c and marginal distributions μ and ν , find the transport plan that minimises the total cost. We now turn to the reverse question. This is a natural extension of the preceding chapters. Once the forward problem is understood, one may question whether the observed transport behaviour can be used to recover the cost structure that generated it.

Recall the power grid example from Chapter 2. There, we assumed knowledge of the transmission costs between power stations and households and sought the cheapest way to route electricity. But suppose instead that we observe how electricity is already being transmitted across the grid, and wish to determine what the underlying transmission costs must have been. This is an instance of an inverse problem: rather than computing an optimal plan from known costs, we aim to recover the costs from an observed plan.

In many real-world settings, we have access to data on how resources flow between locations, but the costs governing those flows are not directly observable. If we assume that the observed flows arise from agents minimising their total transport cost, then the observed data can be interpreted as an approximate solution to a Kantorovich problem. The task then reduces to finding the cost matrix that best explains what we see.

This chapter explores an application of this methodology. We review the methodology of Gaskin et al. [2025] who develop a method for recovering trade costs between countries from observed international trade flow data. Their central modelling assumption is that countries, in aggregate, allocate exports and imports in a cost-minimising manner. Under this assumption, the mathematical framework developed in the preceding chapters applies directly.

5.1 Mathematical Formulation

5.1.1 Discrete Optimal Transport

We begin by considering the discrete setting of the optimal transport problem. This is shown in a concise manner in Thorpe [2023, Section 2.2], and this subsection is based on the treatment provided there.

Let $X = \{x_1, x_2, \dots, x_m\}$ and $Y = \{y_1, y_2, \dots, y_n\}$ be finite sets of source and destination locations respectively, $\mu \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Y)$. In this discrete setting, these may be written as

$$\mu = \sum_{i=1}^m \mu_i \delta_{x_i}, \quad \nu = \sum_{j=1}^n \nu_j \delta_{y_j},$$

with

$$\sum_{i=1}^m \mu_i = \sum_{j=1}^n \nu_j = 1.$$

Let c_{ij} denote the cost of transporting one unit of mass from $x_i \in X$ to $y_j \in Y$.

Given a transport plan π , we interpret

$$\pi_{ij} = \pi(x_i, y_j)$$

as the amount of mass transported from x_i to y_j expressed as a fraction of the total mass moved.

The discrete Kantorovich optimal transport problem then becomes the linear programme

$$\min_{\pi \in \Pi(\mu, \nu)} \left\{ \sum_{i=1}^m \sum_{j=1}^n c_{ij} \pi_{ij} \mid \pi_{ij} \geq 0, \sum_{i=1}^m \pi_{ij} = \nu_j, \sum_{j=1}^n \pi_{ij} = \mu_i \right\}.$$

The final two conditions are the marginal constraints. They ensure that the total mass arriving at each destination y_j is ν_j , and that the total mass leaving each source location x_i is μ_i .

5.1.2 Entropy Regularised Optimal Transport

We now consider the entropy regularised optimal transport problem. In the discrete setting, this takes the following form established in Gaskin et al. [2025]. We omit explicit index ranges on the summations for visual clarity, with the understanding that all sums are taken over the appropriate indices as defined in the previous section.

$$\min_{\pi \in \Pi(\mu, \nu)} \left\{ \sum_{i,j} c_{ij} \pi_{ij} + \varepsilon \sum_{i,j} \pi_{ij} (\log(\pi_{ij}) - 1) \right\}.$$

To solve this problem, we form the Lagrangian

$$\mathcal{L}(\pi, \lambda, \eta) = \sum_{i,j} c_{ij} \pi_{ij} + \left\langle \lambda, \sum_j \pi_{ij} - \mu \right\rangle + \left\langle \eta, \sum_i \pi_{ij} - \nu \right\rangle + \varepsilon \sum_{i,j} \pi_{ij} (\log(\pi_{ij}) - 1).$$

Equivalently, this may be written as

$$\begin{aligned} \mathcal{L}(\pi, \lambda, \eta) &= \sum_{i,j} c_{ij} \pi_{ij} + \sum_i \lambda_i \left(\sum_j \pi_{ij} - \mu_i \right) + \sum_j \eta_j \left(\sum_i \pi_{ij} - \nu_j \right) + \varepsilon \sum_{i,j} \pi_{ij} (\log(\pi_{ij}) - 1) \\ &= \sum_{i,j} c_{ij} \pi_{ij} + \sum_{i,j} \lambda_i \pi_{ij} + \sum_{i,j} \eta_j \pi_{ij} - \sum_i \lambda_i \mu_i - \sum_j \eta_j \nu_j + \varepsilon \sum_{i,j} \pi_{ij} (\log(\pi_{ij}) - 1) \\ &= \sum_{i,j} (\pi_{ij} (c_{ij} + \lambda_i + \eta_j) + \varepsilon \pi_{ij} (\log(\pi_{ij}) - 1)) - \sum_i \lambda_i \mu_i - \sum_j \eta_j \nu_j. \end{aligned}$$

Differentiating with respect to π_{ij} gives

$$\frac{\partial \mathcal{L}}{\partial \pi_{ij}} = c_{ij} + \lambda_i + \eta_j + \varepsilon \log(\pi_{ij}).$$

Setting this equal to zero and solving for π_{ij} yields

$$\pi_{ij} = e^{-c_{ij}/\varepsilon} e^{-\lambda_i/\varepsilon} e^{-\eta_j/\varepsilon}.$$

This can be rewritten in matrix form as

$$\Pi = \Psi e^{-C/\varepsilon} \Omega,$$

where

$$\Psi = \text{diag}(e^{-\lambda_1/\varepsilon}, \dots, e^{-\lambda_m/\varepsilon}), \quad M = e^{-C/\varepsilon}, \quad \Omega = \text{diag}(e^{-\eta_1/\varepsilon}, \dots, e^{-\eta_n/\varepsilon}).$$

As shown in Cuturi [2013], entropy-regularised optimal transport holds several advantages over its standard counterpart. Standard OT requires solving a linear programme with computational cost scaling as $O(d^3 \log d)$. Introducing an entropic penalty can reduce this to $O(d^2)$. Beyond computation, there is an intuitive modelling motivation. Unregularised solutions tend to concentrate mass along a small number of routes, effectively allowing a few matchings to monopolise the transport plan. This can produce unrealistic transport plans that ignore the distributional structure. The entropy term discourages such monopolisation, and instead favours plans that spread mass more naturally across the support.

5.1.3 Sinkhorn Algorithm

Now that we know that $\Pi = \Psi M \Omega$, it remains to solve for Ψ and Ω in order to recover the transport plan Π .

This is done iteratively by forcing $\Pi = \Psi M \Omega$ to alternately satisfy the marginal constraints μ and ν , namely

$$\begin{aligned} \sum_{i=1}^m \Pi_{ij} &= \nu_j, \\ \sum_{j=1}^n \Pi_{ij} &= \mu_i. \end{aligned}$$

This iterative procedure is known as the Sinkhorn algorithm. We provide a description of how it is run based on its presentation in Gaskin et al. [2025].

We begin by initialising $\Psi^{(0)}$ to have all entries equal to 1, and then calculate $\Omega^{(1)}$. To do this, we first note that $\Pi_{ij} = \Psi_{ii} M_{ij} \Omega_{jj}$, since Ψ and Ω are diagonal matrices. Using the marginal constraint $\sum_{i=1}^m \Pi_{ij} = \nu_j$, we find the first update is given as

$$\sum_{i=1}^m \Psi_{ii} M_{ij} \Omega_{jj} = \nu_j \implies \Omega_{jj} \sum_{i=1}^m \Psi_{ii} M_{ij} = \nu_j \implies \Omega_{jj} = \frac{\nu_j}{\sum_{i=1}^m \Psi_{ii} M_{ij}}.$$

We now determine $\Psi^{(1)}$. We use the marginal constraint $\sum_{j=1}^n \Psi_{ii} M_{ij} \Omega_{jj} = \mu_i$, which gives

$$\Psi_{ii} \sum_{j=1}^n M_{ij} \Omega_{jj} = \mu_i \implies \Psi_{ii} = \frac{\mu_i}{\sum_{j=1}^n M_{ij} \Omega_{jj}}.$$

Repeating these alternate rescalings yields the Sinkhorn algorithm. We continue in this manner until a numerical stopping criterion is met. The convergence of this procedure is guaranteed by Sinkhorn [1964], in which it is shown that for any strictly positive matrix A , we can find diagonal matrices D_1 and D_2 such that $D_1 A D_2$ has column and row sums equal to 1. It additionally details that we may find these diagonal matrices by iteratively scaling the rows and columns of the matrix to norm 1. As M is guaranteed to be strictly positive, and we are normalising each entry of Ψ and Ω and scaling the rows and columns to satisfy the marginal constraints, we conclude the algorithm must converge.

5.1.4 Deriving the Entropy Regularised Dual Function

We recall that the entropy regularised Lagrangian is given by

$$\mathcal{L}(\pi, \lambda, \eta) = \sum_{i,j} \left(\pi_{ij} (c_{ij} + \lambda_i + \eta_j) + \varepsilon \pi_{ij} (\log(\pi_{ij}) - 1) \right) - \sum_i \lambda_i \mu_i - \sum_j \eta_j \nu_j,$$

and is minimised at $\pi_{ij} = e^{-c_{ij}/\varepsilon} e^{-\lambda_i/\varepsilon} e^{-\eta_j/\varepsilon}$.

We note that $\log(\pi_{ij}) = -\frac{1}{\varepsilon}(\lambda_i + c_{ij} + \eta_j)$, and hence

$$\varepsilon \pi_{ij} (\log(\pi_{ij}) - 1) = -\pi_{ij} (\lambda_i + c_{ij} + \eta_j) - \varepsilon \pi_{ij}.$$

Substituting this back into $\mathcal{L}(\pi, \lambda, \eta)$ yields

$$\inf_{\pi} L(\pi, \lambda, \eta) = -\varepsilon \sum_{i,j} e^{-(\lambda_i + c_{ij} + \eta_j)/\varepsilon} - \sum_i \lambda_i \mu_i - \sum_j \eta_j \nu_j.$$

Hence, defining $\alpha := -\lambda$, $\beta := -\eta$, the dual problem becomes

$$g(\alpha, \beta) = \sup_{\alpha, \beta} \left\{ \langle \alpha, \mu \rangle + \langle \beta, \nu \rangle - \varepsilon \sum_{i,j} e^{(\alpha_i + \beta_j - c_{ij})/\varepsilon} \right\},$$

as shown in Gaskin et al. [2025].

Examining the final term, if $\alpha_i + \beta_j > c_{ij}$, then we find that

$$g(\alpha, \beta) \rightarrow -\infty \quad \text{as } \varepsilon \rightarrow 0.$$

Based upon this observation, we note for α, β to prove optimal, they must satisfy the optimality criteria

$$\alpha_i + \beta_j \leq c_{ij} \quad \text{for all } i, j.$$

Equivalently, if we define a matrix $D \in \mathbb{R}^{m \times n}$ by

$$D_{ij} = \alpha_i + \beta_j,$$

then D must be entrywise less than or equal to the cost matrix.

5.1.5 Inverse Optimal Transport

In inverse optimal transport, we seek to infer the underlying cost matrix C from observed transport plans Π , which we assume to be solutions of the entropy regularised optimal transport problem.

We begin by rewriting $\Pi = \Psi e^{-C/\varepsilon} \Omega$ as $C = \varepsilon(\log \Psi + \log \Omega^T - \log \Pi)$.

As noted by Gaskin et al. [2025], there may exist many different cost matrices that generate the same transport plan. To address this, Gaskin et al. [2025] impose the bound $c_{ij} \in [0, C_{\max}]$, where $C_{\max}$ is an upper bound on the edge costs. We interpret the case $c_{ij} = C_{\max}$ as corresponding to a movement that we wish to make impossible.

Additionally, imposing the condition

$$\Pi_{ij} = 0 \implies c_{ij} = C_{\max}$$

guarantees uniqueness of the solution, which we state without proof. See Gaskin et al. [2025] for more details.

The procedure for estimating C is as follows. One first initialises a cost matrix $\hat{C}$ and then infers the corresponding transport plan $\hat{\Pi}$ via Sinkhorn's algorithm.

A loss function is then computed, given by

$$\|\hat{\Pi}(\hat{C}) - \Pi\|_2^2 + \sum_{(i,j) \in S} (c_{ij} - C_{\max})^2, \quad S := \{(i, j) \mid \Pi_{ij} = 0\},$$

where the second term acts as a penalty for transport plans that assign flow to routes for which no transport has been observed.

Finally, the cost matrix is updated to decrease the value of the loss function and the process is iteratively repeated. Gaskin et al. [2025] train a neural network to iteratively predict a cost matrix, and use gradient-based methods to minimise the loss function. We will not provide many details on the training method or justification, but will mention that a neural network can be thought of as a composition of many transformations that take the form of an affine transformation followed by a non-linear transformation.

5.2 Results and Discussion

Gaskin et al. [2025] apply the inverse optimal transport framework to international trade flow data, treating each country’s exports and imports as marginal distributions and the observed trade volumes as an approximate transport plan. The method recovers a cost matrix whose entries are consistent with known contributors to trade friction, including geographic distance, shared borders, and common trade agreements.

A natural question concerns the sensitivity of the recovered costs to the regularisation parameter ε . Since the inverse procedure recovers C from the regularised transport plan, the inferred costs depend on the analyst’s choice of ε . Different values may yield different cost matrices from the same data, and there is no current universally accepted procedure for selecting the regularisation strength. This is a limitation of the method.

We also examine the central modelling assumption in greater detail. The framework assumes that observed trade flows arise from countries collectively minimising total transport cost. In practice, trade is shaped by tariffs, political relationships, and historical ties that do not reduce to a single cost function. The recovered cost matrix therefore absorbs all such latent factors into a single numerical summary of effective friction between each pair of countries. This is both a strength, since it provides a principled measure of trade friction without modelling each factor separately, and a limitation, since the resulting costs are difficult to interpret causally.

Returning to the power transmission example, the inverse problem would correspond to observing electricity flows across a network and recovering the underlying transmission costs. Unlike trade, power grid costs can often be measured from physical quantities such as line resistance. The inverse framework could therefore be validated in such a setting by comparing inferred costs against known values. Unexpected discrepancies might reveal the influence of factors not captured by the cost function, such as network congestion, and could serve as directions for extending the theory.

More broadly, the inverse optimal transport framework applies wherever observed data can reasonably be modelled as the outcome of an efficient redistribution process. It has also been used in areas such as computer vision and the modelling of biological processes, illustrating the wide range of settings in which optimal transport methods can be applied. In conclusion, this dissertation has provided an exposition into Optimal Transport as both a theoretical and applied mathematical framework.

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